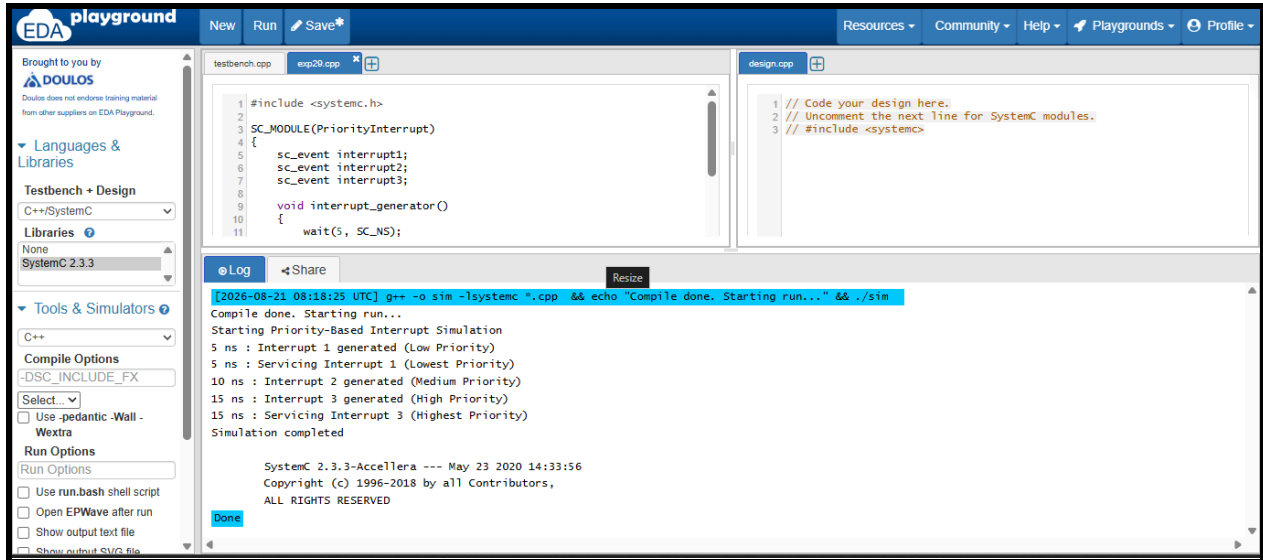




C++ Code:

```
#include <systemc.h>

SC_MODULE(PriorityInterrupt)
{
    sc_event interrupt1;
    sc_event interrupt2;
    sc_event interrupt3;

    void interrupt_generator()
    {
        wait(5, SC_NS);
        cout << sc_time_stamp() << " : Interrupt 1 generated (Low Priority)" << endl;
        interrupt1.notify();

        wait(5, SC_NS);
        cout << sc_time_stamp() << " : Interrupt 2 generated (Medium Priority)" << endl;
        interrupt2.notify();

        wait(5, SC_NS);
        cout << sc_time_stamp() << " : Interrupt 3 generated (High Priority)" << endl;
        interrupt3.notify();
    }

    void interrupt_controller()
    {
        while(true)
        {
            wait(interrupt1 | interrupt2 | interrupt3);
        }
    }
}
```

```

        if(interrupt3.triggered())
        {
            cout << sc_time_stamp()
                << " : Servicing Interrupt 3 (Highest Priority)"
                << endl;
        }
        else if(interrupt2.triggered())
        {
            cout << sc_time_stamp()
                << " : Servicing Interrupt 2 (Medium Priority)"
                << endl;
        }
        else if(interrupt1.triggered())
        {
            cout << sc_time_stamp()
                << " : Servicing Interrupt 1 (Lowest Priority)"
                << endl;
        }

        wait(5, SC_NS);
    }
}

SC_CTOR(PriorityInterrupt)
{
    SC_THREAD(interrupt_generator);
    SC_THREAD(interrupt_controller);
}

};

int sc_main(int argc, char* argv[])
{
    PriorityInterrupt pi("PriorityInterrupt");

    cout << "Starting Priority-Based Interrupt Simulation"
        << endl;

    sc_start(30, SC_NS);

    cout << "Simulation completed"
        << endl;

    return 0;
}

```